



SECUREGATE AI

CONFIDENCE SCORE ANALYSIS

Understanding Distance Score, Confidence Score
and Authorization Decision



1. AUTHORIZED VISITOR EXAMPLE

UPLOADED VISITOR IMAGE



Visitor: Ravi

RESULT GENERATED BY SECUREGATE AI

Metric	Value
Best Match	Ravi
Distance Score	0.0000
Confidence Score	100.00%
Status	AUTHORIZED

WHAT IS DISTANCE SCORE?

Distance Score measures the difference between the uploaded visitor's facial features and the registered resident's facial features.

FaceNet converts every face into a numerical embedding vector.

The system calculates:

$$\text{Distance} = || \text{Embedding}_{\text{visitor}} - \text{Embedding}_{\text{resident}} ||$$

- Smaller Distance → Faces are very similar
- Larger Distance → Faces are different

For Ravi:



Both embeddings are identical, therefore
Distance Score = 0.0000

WHAT IS CONFIDENCE SCORE?

Confidence Score is a user-friendly percentage generated from the distance score.

The formula used is:

$$\text{Confidence} = \frac{(2 - \text{Distance})}{2} \times 100$$

For Ravi:

Distance = 0.0000 → Confidence = 100%

WHY ACCESS WAS AUTHORIZED?

The project uses the following rule:

```
if confidence >= 70:  
  AUTHORIZED  
else:  
  UNAUTHORIZED
```

Condition	Result
Confidence = 100%	Above Threshold
Access Decision	AUTHORIZED

ACCESS GRANTED



EXPLANATION

Since the uploaded image belongs to Ravi and the database also contains Ravi's registered face, the facial embeddings match perfectly.

- ✓ Distance Score becomes 0.0000
- ✓ Confidence Score becomes 100%
- ✓ System grants **AUTHORIZED ACCESS**



CONCLUSION: A lower distance score indicates higher facial similarity.
A higher confidence score leads to authorized access.
SecureGate AI ensures only verified residents are granted entry.





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2. UNAUTHORIZED VISITOR EXAMPLE

UPLOADED VISITOR IMAGE



Visitor: Unknown Person
(Not Registered)

RESULT GENERATED BY SECUREGATE AI

Metric	Value
Best Match	Priya
Distance Score	1.1698
Confidence Score	41.51%
Status	UNAUTHORIZED

WHAT HAPPENED?

The uploaded face does not belong to any registered resident.

SecureGate AI compares the uploaded face against all registered residents and identifies the closest match.

In this example:

- Uploaded Visitor → **Unknown Person**
- Closest Resident → **Priya**

Although Priya is the closest match,
the facial similarity is still very low.

UNDERSTANDING DISTANCE SCORE

Distance Score measures the difference between two facial embeddings generated by FaceNet.

$$\text{Distance} = || \text{Embedding}_{\text{visitor}} - \text{Embedding}_{\text{resident}} ||$$

Distance Interpretation:

Distance Score	Meaning
0.0	Exact Match
0.2 - 0.5	Strong Match
0.6 - 0.8	Moderate Match
Above 1.0	Different Person

For this visitor:

Distance Score = 1.1698

This indicates that the uploaded face is significantly
different from Priya's registered face.

UNDERSTANDING CONFIDENCE SCORE

Confidence Score is calculated from the Distance Score.

$$\text{Confidence} = \frac{(2 - \text{Distance})}{2} \times 100$$

For this visitor:

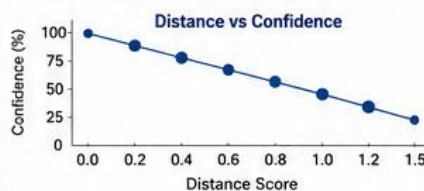
Distance = 1.1698

Confidence = 41.51%

RELATIONSHIP BETWEEN DISTANCE AND CONFIDENCE

Distance ↓ → Confidence ↑

Distance ↑ → Confidence ↓



Distance	Confidence
0.0	100%
0.2	90%
0.4	80%
0.6	70%
0.8	60%
1.0	50%
1.2	40%
1.5	25%

WHY ACCESS WAS DENIED?

The project uses the following rule:

```
if confidence >= 70:  
    AUTHORIZED  
else:  
    UNAUTHORIZED
```

Condition	Result
Confidence = 41.51%	Below Threshold
Access Decision	UNAUTHORIZED



EXPLANATION

Since the uploaded face does not belong to any registered resident:

- ✓ Distance Score remains high (1.1698)
- ✓ Confidence Score remains low (41.51%)
- ✓ Confidence is below the required threshold (70%)
- ✓ System correctly denies access
- ✓ Visitor is classified as **UNAUTHORIZED**



**ACCESS
DENIED**



CONCLUSION

- Distance Score is the actual FaceNet similarity measurement.
- Confidence Score is a user-friendly percentage derived from the Distance Score.
- Authorization Decision is based on the confidence threshold.

- ✓ Confidence ≥ 70% → **AUTHORIZED**
- ✗ Confidence < 70% → **UNAUTHORIZED**

SecureGate AI ensures that only verified residents receive access permission.



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